Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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07/07/2026

Session	2024-25 (ODD)	Course Name	Web Technology Lab
Semester	3	Course Code	23CT1301
Roll No	22	Name of Student	Sanskruti Kharapkar

Practical Number	5
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	**To design and develop an interactive calculator using HTML, CSS and JavaScript that performs arithmetic operations and responds to mouse clicks and keyboard events.**
Problem Definition	**Build an interactive calculator that responds to button clicks and keyboard input, and dynamically updates the result using JavaScript event handling.** **Accept numbers from buttons** **Perform + − × ÷** **Update the display dynamically** **Support keyboard input** **Include a Clear (C) function**

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Theory (100 words)	**An event in JavaScript is an action performed by the user or browser, such as clicking a button or pressing a keyboard key. Event handling allows JavaScript to respond to these actions. The `addEventListener()` method is used to attach an event to an HTML element. In this practical, click events are used for calculator buttons and keyboard events are used to accept numeric and operator input. JavaScript accesses the calculator display using the DOM and dynamically changes its value. Conditional statements such as `if` and `else if` are used to identify arithmetic operators. Functions are used to perform calculations and clear the display.**
Procedure and Execution (100 Words)	Step for Implementation: **1. Create an HTML file containing a calculator display and buttons for numbers and operators.** **2. Add CSS to arrange and style the calculator buttons.** **3. Create JavaScript variables to access the display and buttons.** **4. Use `addEventListener()` to detect button clicks.** **5. Create functions to enter numbers, operators and perform calculations.** **6. Use the DOM to dynamically update the display.** **7. Add a Clear button to reset the calculator.** **8. Add a keyboard event using `keydown` to accept numbers and operators from the keyboard.** **9. Test addition, subtraction, multiplication, division and clear operations in the browser.**

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Code: **Javascript code:**

```
let display = document.getElementById("display");

let numbers = document.querySelectorAll(".number");
let operators = document.querySelectorAll(".operator");

numbers.forEach(function(button) {
    button.addEventListener("click", function() {
        display.value += button.innerText;
    });
});

operators.forEach(function(button) {
    button.addEventListener("click", function() {
        display.value += button.innerText;
    });
});

document.getElementById("clear").addEventListener("click
", function() {
    display.value = "";
});

document.getElementById("equal").addEventListener("click
", function() {
    try {
        display.value = eval(display.value);
    } catch {
        display.value = "Error";
    }
});

document.addEventListener("keydown", function(event) {

    if (
        (event.key >= "0" && event.key <= "9") ||
        event.key == "+" ||
```

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```
        event.key == "-" ||
        event.key == "*" ||
        event.key == "/"
    ) {
        display.value += event.key;
    }

    else if (event.key == "Enter") {
        try {
            display.value = eval(display.value);
        } catch {
            display.value = "Error";
        }
    }

    else if (event.key == "Escape") {
        display.value = "";
    }
});
```

HTML CODE :

```
<!DOCTYPE html>
<html>
<head>
    <title>Interactive Calculator</title>
    <link rel="stylesheet" href="icon.css">
</head>

<body>

    <div class="calculator">

        <input type="text" id="display" readonly>

        <div class="buttons">
```

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```
            <button class="number">7</button>
            <button class="number">8</button>
            <button class="number">9</button>
            <button class="operator">/</button>

            <button class="number">4</button>
            <button class="number">5</button>
            <button class="number">6</button>
            <button class="operator">*</button>

            <button class="number">1</button>
            <button class="number">2</button>
            <button class="number">3</button>
            <button class="operator">-</button>

            <button id="clear">C</button>
            <button class="number">0</button>
            <button id="equal">=</button>
            <button class="operator">+</button>

        </div>

    </div>

<script src="calculator.js"></script>

</body>
</html>
```

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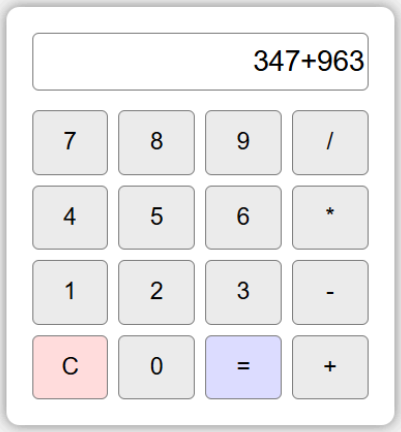
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Output:



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Output Analysis	**The developed calculator successfully responds to both mouse and keyboard events. The `click` event is used for calculator buttons, while the `keydown` event accepts keyboard input. The DOM is used to access and update the display dynamically. Arithmetic expressions are evaluated when the `=` button or Enter key is pressed. The Clear button removes the current expression from the display. Thus, the practical demonstrates JavaScript events, `addEventListener()`, DOM manipulation, functions, conditional statements and basic arithmetic operations.**
Link of student Github profile where lab assignment has been uploaded	Sanskruti2007-code/web-Technology-Practicals

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Conclusion	**The interactive calculator was successfully designed using HTML, CSS and JavaScript. JavaScript event handling was implemented using `addEventListener()` for both mouse clicks and keyboard input. The practical helped demonstrate DOM manipulation, functions, conditional statements and dynamic updating of web page content. The calculator successfully performs basic arithmetic operations and provides a Clear function, making the webpage interactive and user-friendly.**
Plag Report (Similarity index < 12%)	**9%**
Date	**21/09/2026**